Suitable Network Topology: Mesh Topology

1. Speed

- In a mesh setup, game servers across different regions (e.g., Mumbai, Singapore, Europe data centers) are interconnected with **multiple direct paths** to each other.
- If one path is congested or slow, data can automatically reroute through another available path — this is called **dynamic routing**.
- For a fast-paced game like PUBG/Free Fire, this means player actions (shooting, movement, loot pickups) sync faster across servers with minimal lag, since the network always picks the quickest available route rather than being stuck with one fixed path.

2. **Reliability**

- This is mesh's biggest strength: **no single point of failure**.
- If one server or one connection link goes down (hardware failure, cable cut, DDoS attack, etc.), traffic instantly reroutes through other interconnected servers — players don't get disconnected or face a total server crash.
- For competitive multiplayer games, this reliability is critical because even a few seconds of downtime during a match (especially in tournaments) can ruin the player experience and company reputation.
- This also helps with **fault isolation** — if one node is compromised or attacked, admins can isolate it without shutting down the whole network.

3. **Cost**

- The tradeoff is **high cost**:

- Each server needs a dedicated connection to every other server (in a full mesh, this is n(n-1)/2 connections for "n" servers) — this means more cabling, more network interface hardware, and more bandwidth purchased.
- Requires **complex network management** — skilled network engineers, monitoring tools, and redundancy protocols (like BGP routing) to keep it optimized.
- Scaling up (adding more regional servers as your player base grows) also gets expensive quickly since every new server must connect to all existing ones (or at least a partial mesh).